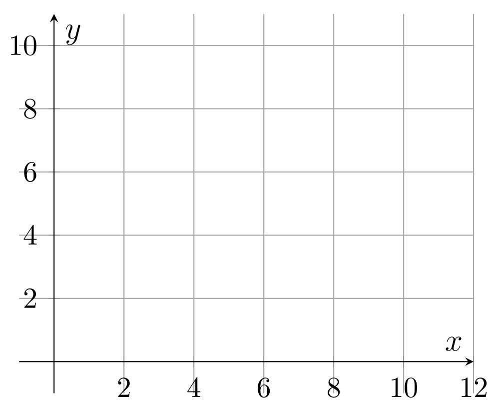
Distance and Midpoint on the Coordinate Plane

Directions

- The **distance** between two points is the hypotenuse of the right triangle whose legs are the horizontal and vertical gaps: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- The **midpoint** is the pair of averages: $M = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$.
- Give every midpoint as an ordered pair. Give distances exactly where they come out whole, and otherwise to two decimal places.
- Two problems include a blank grid. Plotting the points is optional — it is there so you can check that your answer looks right.

1. Find the distance between $A(2, 3)$ and $B(10, 9)$.



Answer: ________________ units

2. Find the midpoint of the segment joining $C(-4, 7)$ and $D(6, 1)$. Give your answer as an ordered pair.

Answer: ________________

3. Find the distance between $P(-5, 2)$ and $Q(7, -3)$.

Answer: ________________ units

4. The point $M(3, 5)$ is the midpoint of segment $\overline{AB}$, and one endpoint is $A(-1, 2)$. Find the coordinates of the other endpoint B.

$B =$ (________, ________)

5. On a city map each unit is one block. A park entrance sits at $(2, 1)$ and a library sits at $(14, 10)$. Find the straight-line distance from the park entrance to the library, in blocks.

Answer: ________________

6. A segment has endpoints $E(-6, 3)$ and $F(2, -5)$.

(a) Find the midpoint of $\overline{EF}$, as an ordered pair.

Answer: ________________

(b) Explain why averaging the two x-coordinates and the two y-coordinates gives a point that lies *on* the segment, rather than merely somewhere between the two x-values.

7. A segment has endpoints $R(-3, -1)$ and $S(9, 4)$.

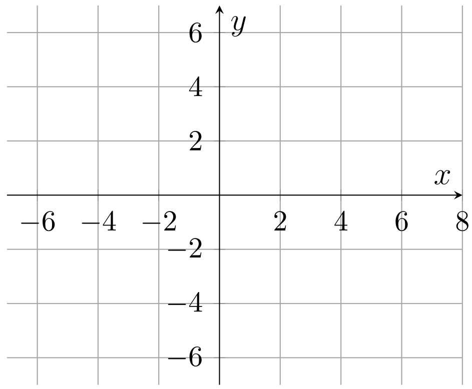
(a) Find the length RS.

Answer: ________________ units

(b) Find the midpoint of $\overline{RS}$, as an ordered pair.

Answer: ________________

8. A quadrilateral has vertices $W(-4,1)$, $X(0,5)$, $Y(6,-1)$ and $Z(2,-5)$, and its diagonals are $\overline{WY}$ and $\overline{XZ}$.



(a) Find the midpoint of $\overline{WY}$, as an ordered pair.

Answer: ________________

(b) Find the midpoint of $\overline{XZ}$, as an ordered pair.

Answer: ________________

(c) Taken together, what do those two answers tell you about this quadrilateral? Explain.